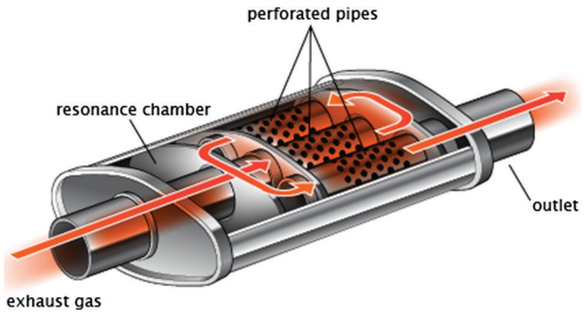


Figure 6.12.3(c) Resonance Type Muffler

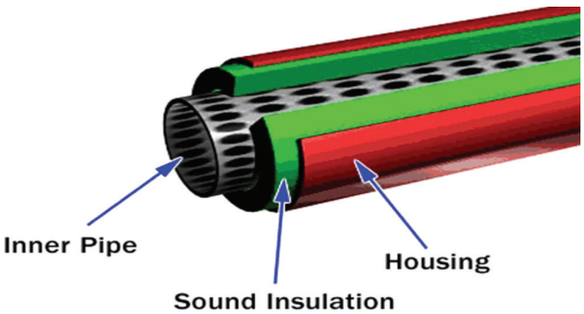


Figure 6.12.3(d) Absorber Type

6.12.3(e) Combined absorber & resonance type

In the combined type mufflers in addition to the absorber material resonators are also fixed on the exhaust path. In this type of mufflers both heat and vibrations are absorbed. Fig 6.12.3(e) shows the Combined absorber & resonance type.

6.13 CATALYTIC CONVERTORS

The exhaust gases coming out from the exhaust manifold are highly toxic to human beings and plants. The catalytic convertor fitted with the exhaust pipe could reduce the harmful emissions such as unburned hydrocarbon, nitrogen oxides and carbon monoxide emissions. In catalytic

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Combined resonance and absorber type muffler: 1. Inlet; 2. Noise absorbant; 3. Access port; 4. Outlet; 5. Resonant

Figure 6.12.3(e) Combined Resonence and Absorber Type

convertors platinum, palladium and rhodium chemicals are used as catalysts which are coated in the form of honey comb like structures and being used in vehicles. A secondary passage is also made to supply oxygen into the convertor. When the toxic gases (such as hydrocarbon and carbon monoxide) are passed into the catalytic convertors, they are converted into water and carbon dioxide. Three way catalytic convertors reduce the nitrogen oxides also by the way of reduction action to nitrogen and oxygen. Fig 6.13 shows the catalytic convertor.

6.14 ENGINE TUNE-UP PROCEDURE

Checking the engine components and adjusting the components for better engine's performance is called as engine tuning. Tuning the engine for better performance increases the life of the engine. The flow chart shown in Figure 6.14 presents the engine tuning procedure on various components of the engine.

To improve the engine's performance (efficiency) the following components must be maintained in good condition.

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